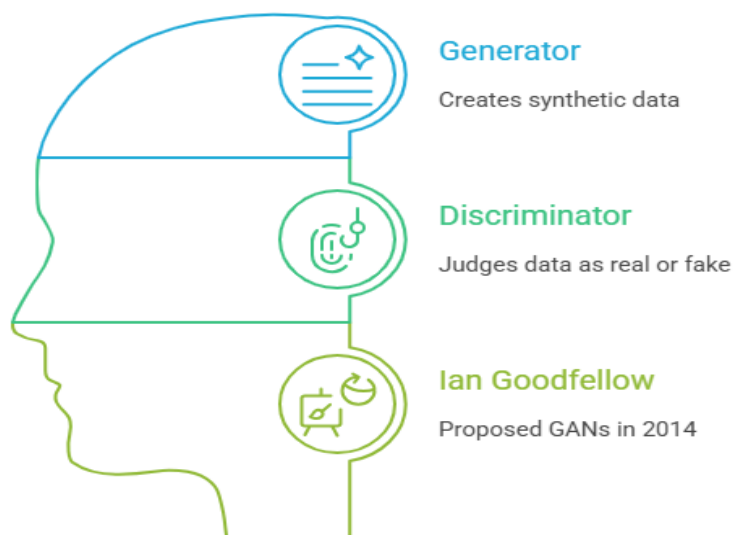


Generative Adversarial Networks (GANs)

Overview

GANs (Generative Adversarial Networks) are generative models — meaning they learn the underlying data distribution and can generate new, realistic samples (like images, music, or text).

- Proposed by Ian Goodfellow in 2014.
- GANs consist of two networks trained together in a game-like setup:
 - **Generator (G)** → Creates synthetic (fake) data.
 - **Discriminator (D)** → Judges whether input data is real (from dataset) or fake (from Generator).

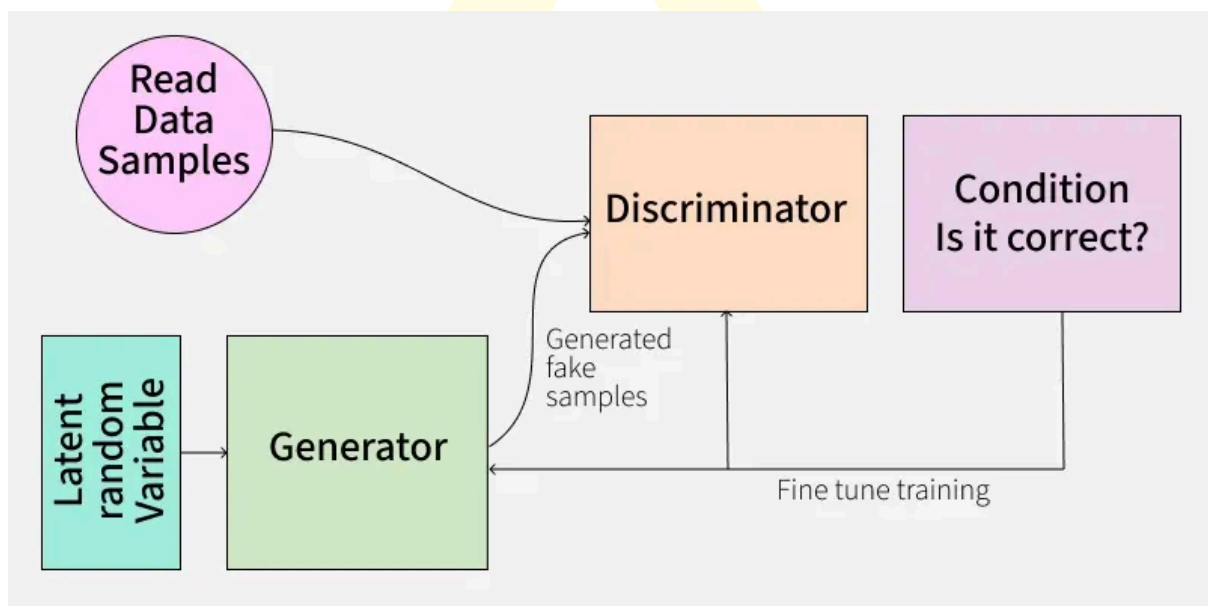


Key idea: Instead of explicitly telling the model what the data distribution looks like, GANs learn it by competition.

Analogy:

- Imagine a counterfeiter (Generator) making fake currency.
- A police officer (Discriminator) tries to detect fake money.
- As training goes on:
 - The counterfeiter learns to make better fakes.
 - The police get better at detecting fakes.
- Eventually → the fakes become so realistic that even the police struggle to differentiate.

GAN Architecture



♦ Generator (G)

- **Input:** A random noise vector $z \sim p_z(z)$, usually sampled from a Gaussian or uniform distribution.
- **Output:** A synthetic sample (image, text, sound).

- **Objective:** Generate data so realistic that the Discriminator believes it is real.

Function:

$$G(z; \theta_g) \rightarrow x^f$$

where θ_g are Generator parameters.

♦ Discriminator (D)

- **Input:** Either a real sample (from dataset) or a fake sample (from Generator).
- **Output:** A probability $D(x)$ indicating if the sample is real (close to 1) or fake (close to 0).

Function:

$D(x; \theta_d) \rightarrow [0, 1]$, where θ_d are Discriminator parameters.

♦ Training as a Minimax Game

The GAN is trained as a two-player minimax game:

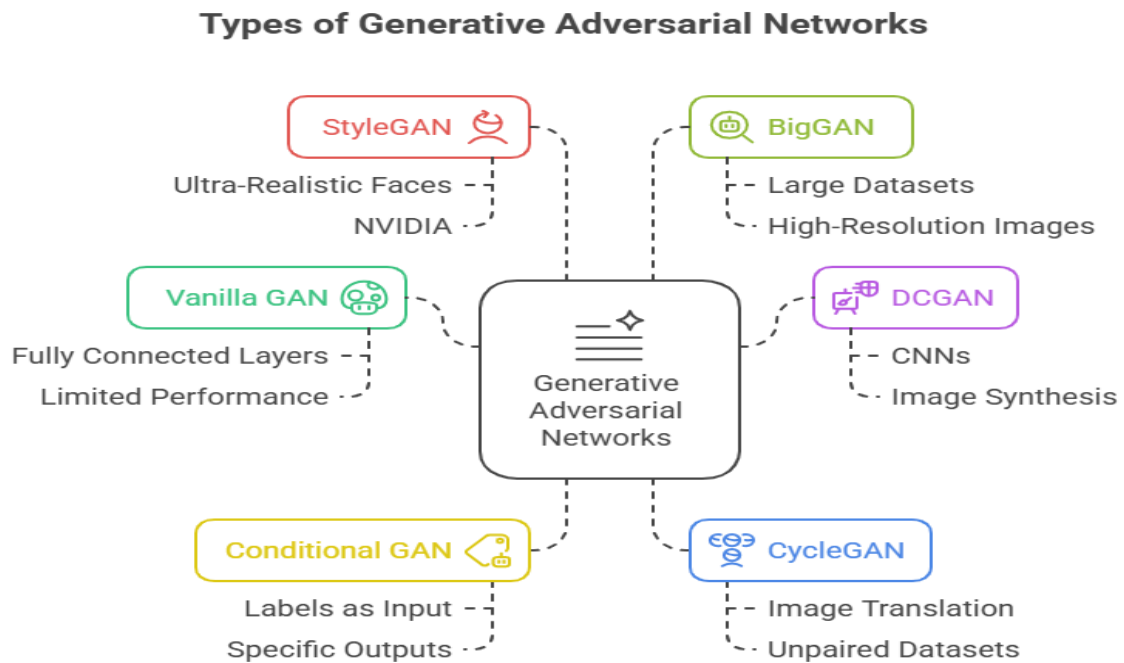
$$\min_D \max_G V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))]$$

- The Discriminator tries to maximize this objective (correctly classify real vs fake).
- The Generator tries to minimize this objective (make fake samples that fool D).

The training reaches an equilibrium when:

- The generator produces samples indistinguishable from real ones.
- Discriminator outputs ~ 0.5 for both real and fake (can't tell the difference).

Types of GANs



1. Vanilla GAN

- The original version.
- Uses fully connected layers.
- Limited performance on complex data like images.

2. DCGAN (Deep Convolutional GAN)

- Replaces fully connected layers with CNNs.
- Generates much better images.
- Widely used for image synthesis.

3. Conditional GAN (cGAN)

- Adds labels as extra input.
- Example: Generate a digit "7" if asked.
- Function:
 $G(z|y), D(x|y)$

4. CycleGAN

- Learn image-to-image translation without paired datasets.
- Example: Convert photos of horses → zebras, summer → winter scenes.

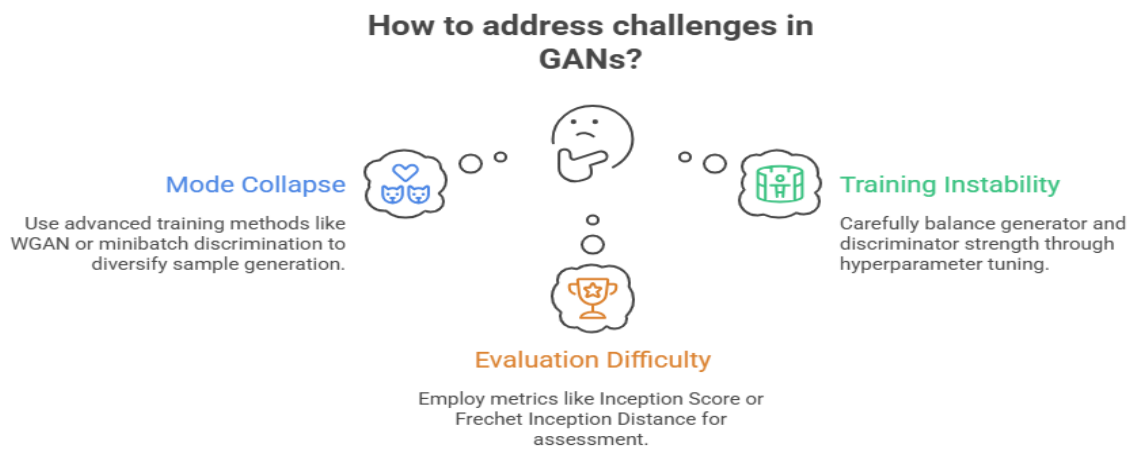
5. StyleGAN

- Introduced by NVIDIA.
- Produces ultra-realistic faces (used in ThisPersonDoesNotExist.com).

6. BigGAN

- Trained on large-scale datasets like ImageNet.
- Can generate high-resolution, diverse images.

Challenges in GANs



◆ Mode Collapse

- Generator produces a limited variety of samples (e.g., only one type of cat).
- **Solution:** Use advanced training methods (WGAN, minibatch discrimination).

◆ Training Instability

- GAN training is very delicate.
- If D becomes too strong, G can't learn. If G becomes too strong, D fails.
- Requires careful balancing and hyperparameter tuning.

◆ Evaluation Difficulty

- Unlike classification (accuracy), GANs don't have a direct metric.
- Common metrics:
 - Inception Score (IS)
 - Frechet Inception Distance (FID)

Applications of GANs

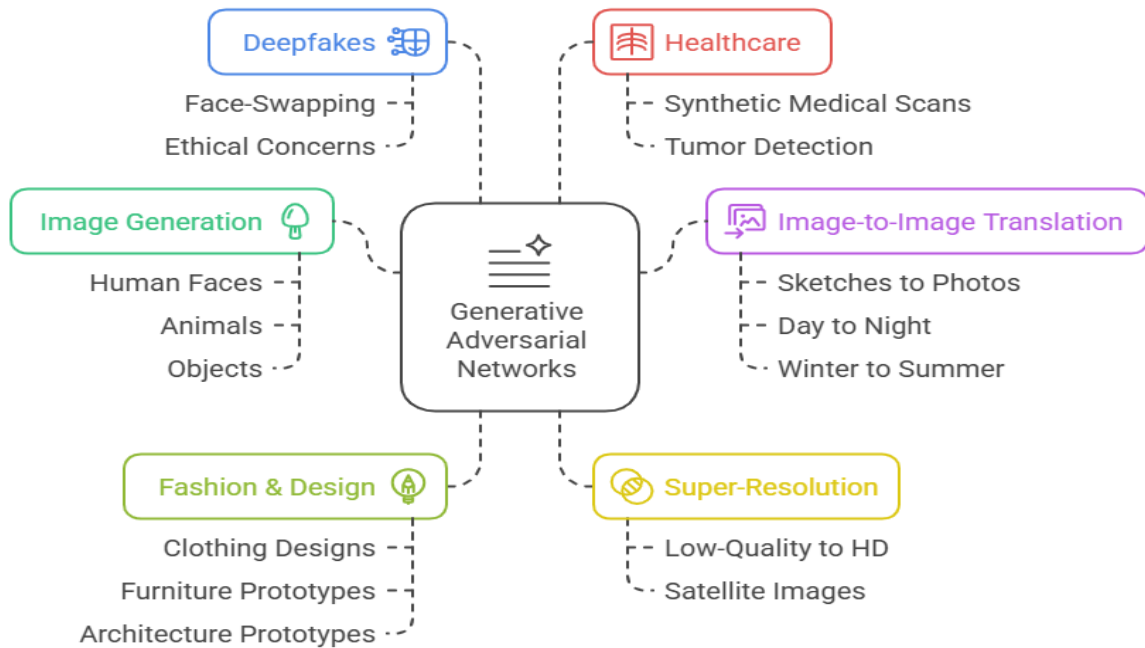


Image Generation

- Generate realistic human faces, animals, or objects.

Image-to-Image Translation

- Convert sketches → photos.
- Day → night, winter → summer.

Super-Resolution

- Turn low-quality images into HD images.
- Example: Enhance satellite images.

Deepfakes

- Face-swapping in videos.

- Used in entertainment (but also raises ethical concerns).

Healthcare

- Generate synthetic medical scans to train models when data is scarce.
- Example: Generate MRI images to improve tumor detection AI.

Fashion & Design

- Generate clothing designs, furniture, or even architecture prototypes.

Real-World Examples

- This Person Does Not Exist → Generates hyper-realistic fake faces using StyleGAN.
- NVIDIA GauGAN → Converts doodles into realistic landscapes.
- DeepMind's BigGAN → High-resolution image synthesis on ImageNet.
- Adobe Photoshop Neural Filters → Uses GANs for editing faces (e.g., age, expression).

Key Takeaways

- GANs are powerful generative models based on adversarial training.
- Consist of two competing networks (Generator vs Discriminator).
- Variants like cGAN, CycleGAN, and StyleGAN extend GANs to many domains.
- Despite challenges (mode collapse, instability), GANs are behind many AI breakthroughs in vision, art, and healthcare.